2D ARRAYS



PREVIOUS LECTURE

- Basics of Arrays
- Declaring Arrays
- Initializing Arrays
- Accessing Arrays
- Processing Arrays

CODE FOR AVERAGE OF AN ARRAY

```
#include <iostream>
using namespace std;

int main() {
    int size;

    cout << "Enter number of elements: ";
    cin >> size;

    int arr[size];   // array declaration
    int sum = 0;

    // Input elements
    cout << "Enter elements:\n";
    for (int i = 0; i < size; i++) {
        cin >> arr[i];
        sum += arr[i];   // add each element to sum
    }

    // Calculate average
    double average = (double)sum / size;

    // Output result
    cout << "Sum = " << sum << endl;
    cout << "Average = " << average << endl;

    return 0;
}
```

MIN AND MAX OF AN ARRAY

```
#include <iostream>
using namespace std;

int main() {
    int size;

    cout << "Enter number of elements: ";
    cin >> size;

    int arr[size];

    // Input elements
    cout << "Enter elements:\n";
    for (int i = 0; i < size; i++) {
        cin >> arr[i];
    }
```

```
    // Assume first element is min and max
    int min = arr[0];
    int max = arr[0];

    // Find min and max
    for (int i = 1; i < size; i++) {
        if (arr[i] < min) {
            min = arr[i];
        }
        if (arr[i] > max) {
            max = arr[i];
        }
    }

    // Output
    cout << "Minimum = " << min << endl;
    cout << "Maximum = " << max << endl;

    return 0;
}
```

OUTLINE

Two Dimensional Arrays

String class in C++

C Style Strings

Outline

Two Dimensional Arrays

String class in C++

C Style Strings

TWO DIMENSIONAL ARRAYS

- A two-dimensional array is like several identical arrays put together. It is useful for storing multiple sets of data.

  - In C++, you can create an *array* of an array known as multi-dimensional array. For example:

```
int arr[3][4];
```

  - Here, **arr** is a two dimensional array. It can hold a maximum of 12 elements.
  - We can think this array as table with 3 rows and each row has 4 columns.

TWO DIMENSIONAL ARRAYS

	Column 0	Column 1	Column 2	Column 3
Row 0	arr[0][0]	arr[0][1]	arr[0][2]	arr[0][3]
Row 1	arr[1][0]	arr[1][1]	arr[1][2]	arr[1][3]
Row 2	arr[2][0]	arr[2][1]	arr[2][2]	arr[2][3]

DECLARING AND INITIALIZINGTWO DIMENSIONAL ARRAYS

- int arr[4][4];
- Visualized as:

- arr[0][0] = 21;

DECLARING AND INITIALIZINGTWO DIMENSIONAL ARRAYS

- int arr[4][4];
- Visualized as:

21			

- arr[0][0] = 21;
- arr[1][2] = 36;

DECLARING AND INITIALIZING TWO DIMENSIONAL ARRAYS

- int arr[4][4];
- Visualized as:

21			
		36	

- arr[0][0] = 21;
- arr[1][2] = 36;

INITIALIZING AND ACCESSING TWO DIMENSIONAL ARRAYS

- int arr[2][3] = { 1,2,3,4,5,6};
- Better way
  - int arr[2][3] = {{9,2,3},{6,9,11}};

	Col 0	Col 1	Col 2
Row 0	9	2	3
Row 1	6	9	11

- cout<<arr[0][2];
- This statement will display
- cout<<arr[1][2];
- This statement will display

PROCESSING ARRAYS

```
#include <iostream>
using namespace std;

int main()
{
    int test[3][2] =
    {
        {2, -5},
        {4, 0},
        {9, 1}
    };

    // Accessing two dimensional array using
    // nested for loops
    for(int i = 0; i < 3; ++i)
    {
        for(int j = 0; j < 2; ++j)
        {
            cout<< "test[" << i << "][" << j << "] = " << test[i][j] << endl;
        }
    }

    return 0;
}
```

CONCLUSION

Declaring, Accessing, and processing Two Dimensional Arrays.

NEXT LECTURE

Understanding Basic Functions of Strings